

REPRODUCIBILITY_STATEMENT.md

Deliverable D-S1 — Reproducibility statement

Project: AMGCR Earthquake Research

Scope: California IRIS / EarthScope FDSN pilot (acquisition → feature matrix)

Date: 29 July 2026

This document describes how to reproduce the **completed certificate-track California pilot workflow** documented in the repository. It does **not** cover EarthESND K-NET scientific reproduction or future European pilots.

Related:

[DATA_AVAILABILITY_STATEMENT.md](#) · [IRIS_DATASET_REPORT.md](#) · [FINAL_DELIVERABLE_AUDIT.md](#)

1. What this workflow reproduces

The pilot chain implemented in `scripts/download/` and `scripts/analysis/` produces:

1. Event manifest and raw MiniSEED under `data/` (when `download` is run)
2. Phase A.1 EDA — summaries, inspection CSV, four EDA figures
3. Phase A.2 Signal analysis — per-event metrics JSON, eight four-panel figures, summary table
4. Phase B.1 Preprocessing — frozen config, per-event stage NPZ, eight comparison figures, QC table
5. Phase B.2 Feature engineering — **8×18** feature matrix, correlation table, four feature figures

Narrative interpretation lives in `docs/EDA_REPORT.md` through `docs/RESULTS_AND_DISCUSSION.md` and in `reports/Research_Proposal_v1.md`. Those reports **describe** outputs; they do not re-run computations.

The **EarthESND reference implementation** in `src/` is separate (88 tests when the full reference tree is installed). **California pilot results were not produced with EarthESND modules** (`docs/RESULTS_AND_DISCUSSION.md`).

2. Software environment

Requirement	Source
Python	≥ 3.11 (pyproject.toml)
Core packages	ObsPy ≥ 1.5, NumPy, Pandas, Matplotlib, SciPy, PyYAML (pyproject.toml)
Optional (reference track only)	pytest ≥ 9.1 (pyproject.toml [project.optional-dependencies] dev)

Suggested setup (from repository root):

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -e ".[dev]"
```

There is **no** separate requirements.txt; dependencies are declared in pyproject.toml.

Network: FDSN download and optional instrument-response steps require internet access to USGS and EarthScope services.

3. Execution order

Run all commands from the **repository root**. Stages depend on prior outputs as shown.

```
Step 0 Acquisition      → data/raw/iris/, data/manifests/iris_california_pilot_events.csv, logs/iris_california_pilot_summary.json
Step 1 EDA (A.1)       → reports/eda/, reports/tables/event_summary.csv, station_summary.csv, reports/figures/ (4 EDA PNGs)
Step 2 Signal (A.2)    → reports/signal_analysis/, reports/tables/signal_analysis_per_event.csv, reports/figures/ (8 signal PNGs)
Step 3 Preprocess (B.1) → reports/preprocessing/, reports/tables/preprocessing_quality_metrics.csv, reports/figures/ (8 preprocessing PNGs)
Step 4 Features (B.2)  → reports/features/, reports/tables/feature_matrix.csv, feature_correlation_matrix.csv, reports/figures/ (4 feature PNGs)
```

Step 0 — Download (optional if raw data already present)

Documented in [IRIS_DATASET_REPORT.md](#) §4:

```
python scripts/download/download_iris_california_pilot.py --event-count 8 --min-events 5
```

Optional CLI paths: --raw-dir, --manifest-csv (defaults: data/raw/iris/, data/manifests/iris_california_pilot_events.csv). The script writes a JSON

summary to `logs/iris_california_pilot_summary.json` by default (see `scripts/download/download_iris_california_pilot.py`).

Public FDSN pilot — no credentials ([IRIS_DATASET_REPORT.md](#) §4).

Steps 1–4 — Analysis (manifest-linked traces only)

Commands match phase reports:

```
python scripts/analysis/run_california_pilot_eda.py
python scripts/analysis/run_california_signal_analysis.py
python scripts/analysis/run_california_preprocessing.py
python scripts/analysis/run_california_feature_engineering.py
```

Step	Script	Primary doc
1	<code>scripts/analysis/run_california_pilot_eda.py</code>	EDA_REPORT.md
2	<code>scripts/analysis/run_california_signal_analysis.py</code>	SIGNAL_ANALYSIS_REPORT.md
3	<code>scripts/analysis/run_california_preprocessing.py</code>	PREPROCESSING_REPORT.md
4	<code>scripts/analysis/run_california_feature_engineering.py</code>	FEATURE_ENGINEERING_REPORT.md

Preprocessing parameters for the completed run are recorded in `reports/preprocessing/preprocessing_config.json`. Feature engineering reads **filtered** arrays from `reports/preprocessing/<short_id>_stages.npz` and P picks from `reports/signal_analysis/<short_id>_metrics.json` ([FEATURE_ENGINEERING_REPORT.md](#)).

4. Artefacts: included in the repository vs excluded

4.1 Typically **included** (version-controlled in this project)

These paths are part of the documented pilot and are intended to be shared via Git (see repository commits; audit date 2026-07-29):

Category	Paths
Scripts	<code>scripts/download/download_iris_california_pilot.py</code> , <code>scripts/analysis/run_california_*.py</code>

Category	Paths
Manifest	data/manifests/iris_california_pilot_events.csv
Phase documentation	docs/IRIS_DATASET_REPORT.md, docs/EDA_REPORT.md, docs/SIGNAL_ANALYSIS_REPORT.md, docs/PREPROCESSING_REPORT.md, docs/FEATURE_ENGINEERING_REPORT.md, docs/RESULTS_AND_DISCUSSION.md
Tables & summaries	reports/tables/*.csv, reports/eda/*.json, reports/eda/mseed_file_inspection.csv, reports/signal_analysis/*.json, reports/preprocessing/*.json, reports/features/*.json, reports/features/feature_matrix.csv
Intermediate arrays	reports/preprocessing/*_stages.npz, per-event metrics JSON under reports/signal_analysis/ and reports/preprocessing/
Consolidated narrative	reports/Research_Proposal_v1.md

Exact Git tracking may vary by commit; the **manifest and analysis outputs under reports/ (except figures)** were part of the completed pilot deliverable set per [FINAL_DELIVERABLE_AUDIT.md](#).

4.2 Intentionally excluded from Git (.gitignore)

Path / pattern	Reason	How to obtain
data/raw/ (including data/raw/iris/*.mseed)	Raw waveforms are large and immutable inputs; re-fetched from FDSN	Step 0 download script
logs/ (including logs/iris_california_pilot_summary.json)	Local run logs	Created when Step 0 runs
figures/ (matches reports/figures/)	Generated PNG outputs; reproducible from analysis	Steps 1–4

Path / pattern	Reason	How to obtain
	scripts	
data/interim/, data/processed/, data/output/	Reserved for other pipelines	Not used for this pilot's published matrix
.venv/, .env	Local environment and secrets	User-created

Audit finding ([FINAL_DELIVERABLE_AUDIT.md](#) §1, §5): on a **fresh clone**, **24 PNGs** under reports/figures/ and **raw MiniSEED** are **not** present unless regenerated or supplied in a separate archive.

4.3 Local-only nuance (documented, not a separate Git policy)

[EDA_REPORT.md](#) and reports/eda/mseed_file_inspection.csv record **twelve** MiniSEED files on disk at EDA time (**eight** manifest-linked + **four legacy** from an earlier download). **All completed analyses use the eight manifest-linked records only.**

5. Regenerating excluded artefacts

Excluded artefact	Regeneration
Raw MiniSEED	python scripts/download/download_iris_california_pilot.py --event-count 8 --min-events 5
Acquisition JSON log	Same download run → logs/iris_california_pilot_summary.json
EDA figures (4)	python scripts/analysis/run_california_pilot_eda.py
Signal figures (8)	python scripts/analysis/run_california_signal_analysis.py
Preprocessing figures (8)	python scripts/analysis/run_california_preprocessing.py (requires prior signal metrics for consistency with documented chain)
Feature figures (4)	python scripts/analysis/run_california_feature_engineering.py (requires preprocessing NPZ + signal metrics)

Full figure index: reports/Research_Proposal_v1.md Appendix A; filenames listed in phase reports.

6. Reproducibility limitations

These limits are **documented project findings**, not new science:

1. **FDSN variability** — Re-running Step 0 may yield different events or stations if catalog or dataset availability changes; the **published manifest** (8 events, CI.ADO / CI.USC) reflects the **2026-07-29** run ([IRIS_DATASET_REPORT.md](#) §3).
 2. **Station selection** — Fixed distance-ranked candidate list; not uniform network sampling ([IRIS_DATASET_REPORT.md](#) §2.2, §5).
 3. **Origin-aligned windows** — 300 s from catalog origin; no pre-event quiet time for noise ([SIGNAL_ANALYSIS_REPORT.md](#) §1.2).
 4. **Instrument response** — Preprocessing run: response removal **0/8 applied**; filtered traces remain in **counts** ([PREPROCESSING_REPORT.md](#) §4).
 5. **Small sample** — **N = 8** events, single **BHZ** component; statistics are illustrative ([RESULTS_AND_DISCUSSION.md](#)).
 6. **EarthESND** — Reference code in src/ is **not** on the California reproduction path.
 7. **Bit-identical PNGs** — Matplotlib/ObsPy versions may produce minor visual differences; tabular/json outputs should match for the same inputs and manifest.
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7. Verification pointers

After regeneration, compare against frozen summaries (completed pilot):

- reports/eda/dataset_summary.json
- reports/signal_analysis/signal_analysis_summary.json
- reports/preprocessing/preprocessing_summary.json
- reports/features/feature_summary.json

Phase reports and [FINAL_DELIVERABLE_AUDIT.md](#) list expected file counts (e.g. **24** PNGs, **8** manifest events).

8. References (software and methods cited in repository)

Krischer, L., Megies, T., Barsch, R., Beyreuther, M., Lecocq, T., Caudron, C., and Wassermann, J. (2015). ObsPy: A bridge for earthquake science. *Seismological Research Letters*, 86(3), 765–771.

International Federation of Digital Seismograph Networks (FDSN). FDSN web services specification. <https://www.fdsn.org/webservices/>

Project documentation: [IRIS_DATASET_REPORT.md](#), phase analysis reports under docs/, [Research_Proposal_v1.md](#) §12.

Document control: D-S1 reproducibility statement v1.0 — 29 July 2026. No source code modified to produce this file.